# ReadiLY

AI-Powered Oral Reading Diagnostics for Philippine Classrooms

## Business Plan

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Startup QC 2026 â€” Student Category, Squad 3

## 1. Executive Summary

ReadiLY is an offline-first Android application that converts a three-minute oral reading check into a structured diagnostic: words-correct-per-minute (WCPM), accuracy percentage, and a per-word miscue classification (omission, substitution, self-correction), scored entirely on-device. It targets Grades 1-3 public school teachers in the Philippines, where DepEd's own Phil-IRI and CRLA assessments are administered manually, one student at a time, taking 15-20 minutes per learner and leaving most teachers unable to assess a full class within a grading period.

ReadiLY does not replace Phil-IRI or CRLA; it functions as a fast, repeatable, teacher-facing companion that surfaces word-recognition level (Independent/Instructional/Frustration, using Phil-IRI's own thresholds), a rule-based "needs attention" flag, and a per-student trend view, so teachers can act between formal assessment cycles instead of only at them.

At MVP, scoring is English-only: the underlying on-device speech model (a compact, non-hallucinating transducer architecture, not a large generative model) has no viable Tagalog equivalent under a 200MB on-device budget as of this writing, so Filipino-language passages are recordable and reviewable but not yet auto-scored â€” a deliberate, disclosed engineering boundary, not an oversight. Reading comprehension, predictive risk modeling, and multi-language ASEAN scoring are explicitly out of scope for MVP and are named as a dated roadmap, not a current capability.

ReadiLY is pre-pilot and pre-revenue. This plan states clearly what is built, what is validated by external research (not by us yet), and what remains an assumption to be tested in a 1-3 school pilot â€” the credibility of this submission rests on that distinction being honest.

## 2. Why Join Startup QC

ReadiLY addresses a literacy gap Quezon City itself has already quantified and prioritized: the QC CRLA Report (2025-2026) classifies 39.71% of learners as struggling readers, and the QC School Action Plan 2025-2026 names automated remediation of learning gaps as a stated objective. ReadiLY is a direct technical response to that stated city priority, not a generic edtech pitch retrofitted to Quezon City.

What we need from Startup QC specifically: pilot access to 1-3 QC public schools under a research-permit arrangement (DepEd Order No. 16, s. 2017 governs this and typically takes 3-6 months to negotiate â€” a timeline we cannot shorten without institutional backing); mentorship on pricing and unit economics, which this plan now incorporates from the program's own Unit Economics session; and credibility with SDO Quezon City (SDS Dr. Carleen S. Sedilla, CID Chief Dr. Heidee F. Ferrer), which already hosts comparable pilots (Adarna House's Handang Magbasa/WIKAHON, Khan Academy Philippines) funded through the QC LGU Education Affairs Unit rather than national procurement â€” a faster path than a full DepEd Central Office negotiation.

## 3. Problem Statement

The scale of the underlying problem is externally documented, not just internally asserted: the World Bank's April 2024 Learning Poverty Brief puts Philippine learning poverty (10-year-olds unable to read and understand a simple text) at 91% as of 2022, up sharply from 69.5% in 2019 â€” among the highest rates in the East Asia and Pacific region. This is consistent with the Philippine Statistics Authority's 2024 FLEMMS survey, which found

only 70.8% functional literacy among Filipinos aged 10-64 nationally, with wide regional variation (59.3% in Zamboanga Peninsula versus 81.2% in the Cordillera Administrative Region). These are the correct external figures to cite for this problem â€” earlier drafts of this plan referenced only secondary, unverified citations, which we have now replaced with the primary World Bank and PSA sources.

Early-grade reading assessment in Philippine public schools is manual, slow, and non-diagnostic. Teachers administer oral reading checks one student at a time; a typical class of 40+ students makes full-class, per-grading-period assessment impractical within instructional time. The result is not an absence of assessment tools â€” Phil-IRI and CRLA are already mandated â€” but an absence of assessment frequency and turnaround: a teacher who identifies a struggling reader through Phil-IRI at the start of a grading period has no fast, repeatable way to check progress before the next formal cycle.

This is a scale problem, not a tooling-absence problem. The Philippines averages roughly 36 students per teacher nationally, with urban public schools often exceeding 40-50 (versus 8-11.6 in Singapore/Malaysia). Under those ratios, adding more manual assessment is not viable; the constraint is teacher time, and any credible solution has to reduce it, not add to it.

The immediate, addressable version of this problem is: can a teacher get a fast, repeatable read on word-recognition fluency between formal assessment windows, using a device the school already has, without an internet connection, without additional training burden, and without sending any student audio off the device? That is the specific problem ReadiLY's MVP is scoped to solve â€” not full reading-comprehension diagnosis, and not predictive intervention, both of which require data and validation ReadiLY does not yet have.

## 4. Goals & Objectives

Near-term (0-12 months): complete a structured pilot in 1-3 Quezon City public schools (2-3 teachers, 100-150 students, 2-4 weeks of use) to validate that the scoring pipeline produces results teachers trust and act on; secure a Schools Division research permit; and collect the first Filipino/Tagalog children's-speech data needed to scope a Phase 2 bilingual scoring model.

Medium-term (12-24 months): convert pilot evidence into a paid engagement with at least one QC school or division office, and add the two capabilities the pilot is expected to surface as most requested â€” a comprehension-question layer (leveraging the fact that Phil-IRI passages already carry 5-7 comprehension questions we can adapt) and an admin/division-level reporting export.

Longer-term (24+ months): expand to additional Metro Manila divisions using LGU Special Education Fund (SEF) financing precedent, and begin the Tagalog/Cebuano ASR fine-tuning work using academically-licensed corpora (UP-DSP-PLD, CFSC, BK3AT) that our own research has already identified and scoped.

Success metrics we will track from day one of the pilot: assessment time per student (target: under 4 minutes including setup), percentage of a class assessed within one week (target: 100%, versus partial manual sampling today), and teacher-reported trust in the score (measured via post-pilot survey, not assumed).

## 5. Solution & Product

Core loop: a teacher selects a student and a leveled passage (Filipino or English, grade-tagged), the student reads aloud, the app records locally, and an on-device speech recognizer aligned against the passage's known text produces a diagnostic in under 30 seconds â€” WCPM, accuracy percentage, and a per-word miscue map (omission, substitution, self-correction, using a Phil-IRI-consistent lookahead rule so genuine self-corrections aren't miscounted as errors).

Teacher Dashboard: a class summary screen (counts by Independent/Instructional/Frustration band, a coverage bar showing how many students have been assessed, and a rule-based "needs attention" list); and a per-student profile with a WCPM/accuracy trend chart, a miscue-type breakdown, and full session history. The "needs attention" flag is explicitly rule-based â€” latest band is Frustration, or accuracy is trending down over the last three sessions, or the student sits in the bottom decile of the class by WCPM â€” not a predictive model, and the UI is written to say so, because we cannot yet validate a predictive claim.

Speech recognition & Filipino/dialect handling: MVP scoring uses a compact, non-generative on-device transducer model (deliberately not a large seq2seq model, which in our own technical research is shown to hallucinate corrected text over genuine reading errors â€” the single largest accuracy risk for this use case). A grapheme-level normalizer treats common Philippine-English accent substitutions (e.g., p/f, b/v) as non-errors by default. Tagalog scoring is not yet available at MVP because no on-device Tagalog model exists today within our size and memory budget; Filipino passages remain fully usable for practice and teacher review, with scoring explicitly labeled "coming soon" rather than silently absent. Fine-tuning a scoring model for Filipino and other mother-tongue instruction is a named Phase 2 workstream, contingent on academic data-sharing agreements for existing children's-speech corpora.

Offline capability: class rosters, the passage library, recording, scoring, and the full teacher dashboard all function with zero network connectivity â€” this is a hard design constraint, not a fallback mode. Only an optional background sync feature requires internet, and it never transmits audio: only lightweight score/metadata records sync to the cloud, on a periodic schedule, and student identities are minimized to initials plus DepEd Learner Reference Number rather than full names. The app is fully usable without ever creating an account.

Reading comprehension (roadmap, not MVP): not scored at launch. The nearest credible path is adapting the comprehension questions Phil-IRI passages already include, rather than inventing a new comprehension framework â€” this is deferred to the phase after pilot feedback confirms teachers want it enough to justify the added assessment time it would require.

## 6. Proof of Concept

A working prototype of the end-to-end assessment loop exists: recording capture, on-device scoring, per-word alignment against the reference passage, and error classification. Internal testing has focused on verifying the pipeline does not hallucinate: an automated test asserts that a verbatim reading scores at or above 90% accuracy, and that a passage padded with unread text correctly registers those words as omissions rather than silently inflating the score â€” the specific failure mode that rules out large generative speech models for this use case.

What has not yet happened: no classroom pilot, no teacher using the tool with real students, and no data on whether teachers find the diagnostic trustworthy or actionable in practice. This is the immediate purpose of the pilot phase requested through Startup QC, and we are not presenting simulated or internal-only testing as equivalent to classroom validation.

## 7. Market Analysis

Using a bottom-up sizing method rather than a top-down national estimate: the Philippines has an estimated 47,000 public elementary schools and 300,000+ Grades 1-3 teachers nationally (TAM, structurally). Our Serviceable Addressable Market (SAM) is schools within Metro Manila divisions already running literacy-improvement initiatives comparable to Quezon City's â€” an estimated 15-20% of the national base, given existing SEF-funded literacy programs in QC, Cabanatuan, Makati, and Bais City. Our Serviceable Obtainable Market (SOM) for the next 24 months is 1-2 divisions and 50-100 private/LGU-funded schools reachable through direct partnership rather than national procurement.

First Viable Market check (fixed operating cost Ã· contribution per customer): at an illustrative annual operating cost of â‚±600,000-900,000 (small team, cloud hosting at Supabase Pro tier, pilot logistics) and an illustrative flat per-school license of â‚±30,000/year with an estimated 60% contribution margin after hosting and support costs, the minimum viable customer volume is approximately 35-50 schools â€” a number we can name and test for reachability, rather than an unqualified market-share percentage.

Named competitors: Amira Learning (US, cloud-dependent, $20/student/year full suite, r=0.98-0.99 fluency accuracy vs. human raters â€” not offline, not Philippine-localized); Microsoft Reading Progress (already deployed in DepEd divisions including Bais City and Cabanatuan via Microsoft's national enterprise agreement with DepEd â€” a legal/procurement advantage we do not have and must route around via a division-level research permit instead); Wadhwani AI's Vachan Samiksha (India, 30,000+ government schools, asynchronous on-device-then-sync architecture closest to ours); and SoapBox Labs (fully offline child-speech engine, not localized for Philippine languages). Phil-IRI/CRLA themselves are the incumbent manual process we complement rather than replace.

Our differentiation is not "AI" in the abstract â€” Microsoft and Amira already have that. It is: (1) fully offline scoring where Microsoft Reading Progress is Teams/cloud-dependent and known to fail in low-connectivity "Last Mile" schools; (2) a non-hallucinating architecture choice that we can defend technically against the false-positive/false-negative complaints documented against cloud ASR tools in noisy classrooms; and (3) a procurement path (division-level research permit and LGU SEF financing) that does not require the national enterprise agreement Microsoft already holds.

## 8. Business Model & Financial Plan

Revenue model: flat annual per-school licensing rather than per-student pricing, priced to avoid the enrollment-audit friction per-student billing creates with school MOOE budgets. Illustrative starting price: â‚±30,000/school/year, informed by (not validated against) our own research team's recommendation and comparable Philippine edtech per-student pricing (Quipper: â‚±4,000-10,000/student/year; this is materially higher than our flat-fee approach, which is intentional). Government/division-level licensing and NGO/CSR-

sponsored school access are the two channels we will pursue in parallel, sequenced private-school and NGO partnerships first since B2G sales cycles with DepEd/LGUs are measured in months, not weeks.

Unit economics (illustrative, to be replaced with real figures after the pilot): Customer Acquisition Cost is not yet measurable pre-pilot; we will track it from the first paid engagement onward. Contribution margin per school-year is estimated at approximately â‚±18,000 (60% of the â‚±30,000 license) after cloud hosting (~â‚±1,300-6,500/month at pilot-to-division scale, based on Supabase/Neon pricing tiers) and support costs. Our target LTV:CAC ratio is at least 3:1, the standard benchmark taught in our own Unit Economics session â€" we do not yet have the data to confirm we meet it, and we say so rather than assert a number we cannot support.

Cost structure: fixed costs are AI/app development (current team time, unpaid at this stage) and cloud infrastructure; variable costs are teacher onboarding, pilot logistics, and customer support. No churn data exists yet; for planning purposes we use the SaaS-SMB industry benchmark of 3-7% monthly churn until pilot data replaces it, per standard early-stage guidance to be conservative before real data exists.

Break-even is not expected within the pilot phase. It is contingent on reaching the First Viable Market volume above (roughly 39-61 schools depending on our fixed-cost base that year) or securing grant funding (Startup QC's own equity-free grants up to â‚±100,000 for student innovators, DOST, or DTI) to extend runway through the validation period. The illustrative three-year path below is deliberately conservative and treated as a hypothesis to be replaced with real numbers after the pilot, not a forecast we are asking anyone to rely on.

| **Year** | **Paying Schools (cumulative)** | **Revenue (PHP)** | **Fixed Operating Costs (PHP)** | **Contribution Margin (PHP)** | **Net Position (PHP)** |
|---|---|---|---|---|---|
| Year 1 (pilot + early sales) | 5 | 150,000 | 700,000 | 90,000 | (610,000) |
| Year 2 (division expansion) | 25 | 750,000 | 900,000 | 450,000 | (450,000) |
| Year 3 (approaching breakeven) | 60 | 1,800,000 | 1,100,000 | 1,080,000 | (20,000) |

## 9. Team

Team IKIYUH: Yhasmen Nogales (AI & Machine Learning Lead) and Caryl Joy A. Atienza (Full Stack & Product Development Lead), undergraduate students at the Polytechnic University of the Philippines. Documented achievements include a 1st Runner-Up finish at the Agentic AI Hackathon 2025, a DOST 7th Imake.wemake win, and a 1st Runner-Up finish at BPI Datawave 2025.

As a two-person student team, our honest gaps are: an education domain expert with direct DepEd classroom experience, ML/ASR engineering depth beyond our current level, and B2G sales relationships. We have not consulted any named education expert or

academic advisor to date, and we are not claiming otherwise. Our plan to close this gap is a future advisory board recruited through Startup QC mentorship, direct pilot-school teacher partnerships (which double as domain validation), and the academic institutions (UP Diliman's Speech Processing Lab, De La Salle University) that already hold the Filipino children's-speech corpora our Phase 2 ASR roadmap depends on.

## 10. Risk Assessment

Legal/privacy risk: student voice recordings are legally classified as biometric personal information under the Data Privacy Act (RA 10173) and RA 10367, and combined assessment data qualifies as Sensitive Personal Information requiring explicit parental consent â€” not the school's general authority over students. Mitigation: on-device-only audio processing (never transmitted), a standardized consent form aligned to DepEd Order No. 40 s.2012 before any pilot, and a Schools Division-level Outsourcing Agreement (since we are a data processor, not controller) rather than assuming an informal MOA suffices.

IP/content risk: CRLA assessment materials are licensed CC BY-NC 4.0 by DepEd/USAID/RTI International â€” commercial use requires a bespoke license we do not currently have. This is flagged in our own research as a critical blocker and must be resolved (or avoided by using only our own leveled passages) before any commercial pilot references CRLA-branded content.

Technical risk: Filipino/Tagalog ASR accuracy is unproven at our current stage; mitigation is to disclose this limitation honestly rather than imply broader language coverage, and to scope Phase 2 fine-tuning against named, accessible corpora (CFSC, BK3AT) rather than an unspecified future model.

Go-to-market risk: long B2G sales cycles could stall revenue; mitigation is sequencing private-school/NGO partnerships before DepEd/LGU procurement. Team-capacity risk: two students covering ML, product, and sales simultaneously; mitigation is prioritizing mentor/advisor relationships over premature hiring and keeping MVP scope to the smallest version that proves the core loop.

## 11. Social Impact & Conclusion

ReadiLY's impact case rests on a defensible chain: faster, more frequent word-recognition checks let a teacher act on a struggling reader within a grading period instead of only at the next Phil-IRI cycle. We do not claim a quantified literacy-to-workforce or literacy-to-GDP linkage in this plan, because our own research base does not contain a citable statistic supporting one at the specificity level a judge would expect â€” asserting one anyway would be exactly the kind of unsupported claim this plan is trying to avoid. What we can state directly: DepEd's own "No Read, No Pass" policy (DepEd Order No. 45, s.2002) and the ARAL Act (RA 12028) already institutionalize reading proficiency as a gate to grade progression, meaning a tool that helps teachers catch reading gaps earlier is aligned with an existing state mechanism, not a hypothetical one.

ReadiLY is not yet validated at scale, and this plan says so throughout rather than papering over the gap. What we are asking Startup QC for is specific and achievable within a cadetship timeline: pilot access, mentorship on the unit economics and pricing this plan has begun to model, and the institutional credibility to secure a Schools Division research permit faster than we could alone. The next twelve months are about closing exactly the gaps this audit identifies â€” not about claiming they are already closed.

## References

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Internal research: "Legal and Compliance Assessment," "The Landscape of AI-Powered Oral Reading Assessments," "Validation of Automated Oral Reading Fluency Systems," "Technical Feasibility Assessment," "Architecting Offline-First Educational AI in the Philippines," "Strategic Pathway for AI-Driven EdTech Integration in Philippine Public Schools," "Systemic Realities of Early Grade Reading Assessments" (docs/research/, ReadiLY team, 2026).

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